

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech. (CL) Examination

December 2012

CL 425 Soil Erosion & Conservation (Elective - I)

Date: 08.12.2012, Saturday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) What are the needs of conservation ethics? What are the impacts of conservation practices on the environment? [4]
- (b) What are causes of soil erosion? Enlist factors affecting soil erosion. [3]
- Q - 2 (a) (i) How soil erosion affects various fields? [2]
- (ii) How diversion drains will be helpful to reduce the soil erosion? [2]
- (b) List out principles of water erosion control. Discuss each in detail. [4]
- (c) Discuss following methods briefly: [6]
- (i) Contouring (ii) Strip cropping (iii) Crop rotation
- (iv) Stream bank control (v) Mulching (vi) Farm ponds

OR

- Q - 2 (a) (i) Discuss co-relation between development, conservation and quality of water resources affected by soil erosion. [2]
- (ii) Define waterway. Enlist the data required to design it. Also write steps to design it. [2]
- (b) Give detailed classification of mechanical & biological methods of water erosion control. [4]
- (c) Write short notes on Bunding and Terracing methods to control soil erosion. [6]
- Q - 3 (a) Draw a typical profile of Gully showing permanent structure. [4]
- (b) What are the advantages of drainage? Classify drainage with neat sketches. [5]
- (c) Discuss in detail steps involved and methods used for conservation of soil in hilly areas. [5]

OR

- Q - 3 (a) Write the function and limitations of following control structures: [4]
- (i) Drop spillway (ii) Chute (iii) Formless flume (iv) Pipe spillway
- (b) Discuss in detail the role of forests in soil conservation. What precautions and measures should be taken to protect the forests? [5]
- (c) How the growth of gullies can be checked? How reclamation of gullies is carried out? [5]
- Write the steps taken in practice to control or reclaim gullies.

SECTION - II

- Q - 4 (a) Explain mechanics of water erosion with neat sketch. [4]
- (b) Explain how soil erosion is affected by raindrop? [3]

- Q - 5 (a)** Differentiate between following with figures: [6]
 (i) Rill & Gully erosion (ii) Splash & Sheet erosion (iii) Stream & Water fall erosion
(b) Discuss in detail the constants affecting estimation of soil erosion by wind. [6]
(c) Enlist various factors affecting soil erosion by water. Also list out factors affecting classification of land for soil conservation. [2]

OR

- Q - 5 (a)** The row crop is grown for 4 years in rotation having terraces with graded channel. [8]
 Determine the soil loss from an area for data given below:
 Rainfall & runoff erosivity index = 4500
 Soil erodibility factor = 0.027 Mg/ha-year
 Field slope = 12% and length of slope = 150 m
 Cover management factor = 0.16
(b) Elaborate mechanics of wind erosion with neat sketch. [6]
Q - 6 (a) Enlist various factors affecting erosion by wind. Elaborate each of it. [5]
(b) What are the principles of wind erosion control? Explain each in detail. [5]
(c) Briefly explain any four control practices to reduce the soil erosion by wind on dryland cultivated soil. [4]

OR

- Q - 6 (a)** Explain in detail wind forces acting on soil particle with neat diagram. [5]
(b) Give detailed classification of wind erosion control practices. [6]
(c) Briefly explain **any three** methods to control soil erosion by wind. [3]

Criteria for Example [Q-5 (a)]:

(1) Slope stiffness factor (S_s):

$$s < 4 \text{ m long, } S_s = 3.0 (\sin \theta)^{0.8} + 0.56$$

$$s > 4 \text{ m long \& } < 9\%, S_s = 10.8 (\sin \theta) + 0.03$$

$$s > 4 \text{ m long and } s \geq 9\%, S_s = 16.8 (\sin \theta) - 0.50$$

(2) Strip Cropping Factor (P_s):

$P_s = 1.00$ for contouring only or for alternating strips of corn and small grain

= 0.75 for 4-year rotation with 2 years of row crop

= 0.50 with 1 year of row crop

(3) Terrace Sedimentation Factor (P_t):

$P_t = 1.0$ for no terraces

= 0.2 for terraces with graded channel sod outlets

= 0.1 for terraces with underground outlets

(4) Contouring Factor P_c , Maximum Slope Length and Maximum Strip Crop Width for Different Crops:

Land Slope (%)	P_c value	Maximum Slope Length [†] (m)	Maximum Strip-Crop Width (m)
1-2	0.6	120	40
3-5	0.5	90	30
6-8	0.5	60	30
9-12	0.6	36	24
13-16	0.7	24	24
17-20	0.8	18	18
21-25	0.9	15	15

^{*}Factor for farming upslope and downslope is 1.0.

[†]Maximum slope length for strip cropping can be twice that for contouring only.
